Daily Sales Forecasting

Corporación Favorita

A Comparative Modeling Approach

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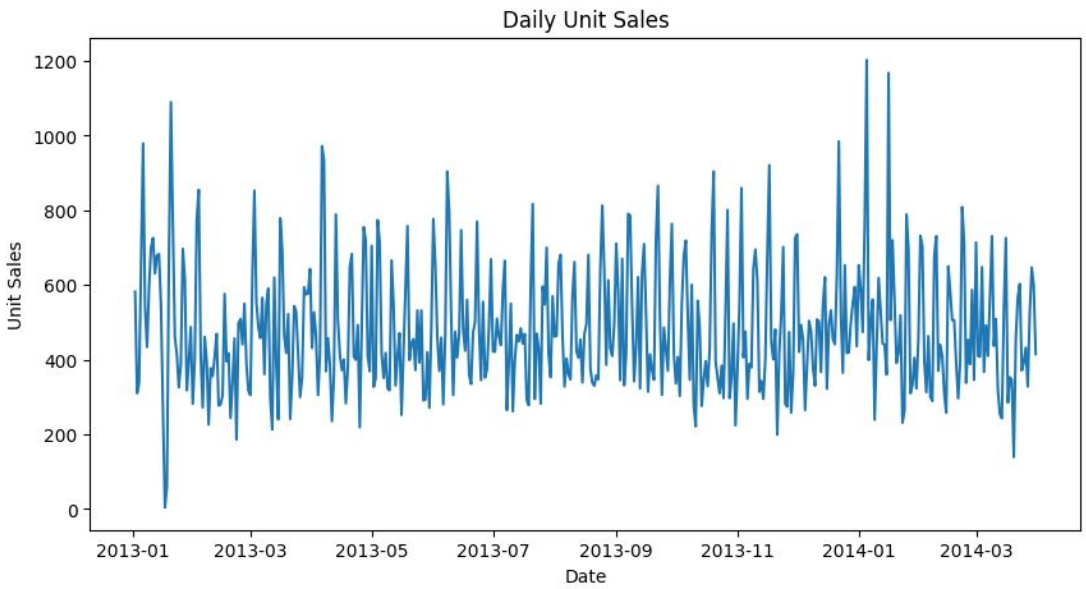
A Time Series Problem



Ordered in time. Observations are not independent

The past influences the future

Future cannot leak into the past

Problem Framing

The Challenge

How much can one year of data tell us?

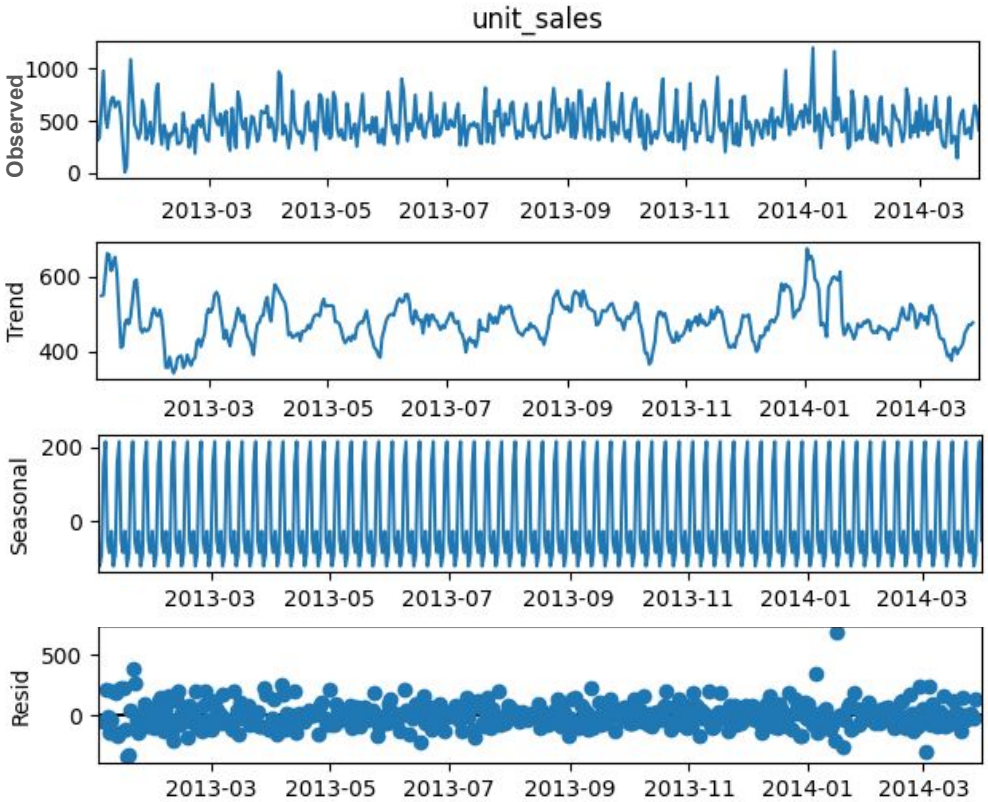
The Data

- 454 daily observations - one year of training
- Single aggregated time series - no store or item breakdown
- Limited external variables - holidays and oil prices only
- Two missing days imputed (Dec 25, 2013 - Jan 1, 2014)

The Goal

- Forecast daily sales for Q1 2014 (Jan 1 → Mar 31)
- Evaluate multiple modeling approaches
- Deploy the best model in an interactive app

What the Data Tells Us Before Any Model



Moderate fluctuations around a stable average . No extreme drift.

No strong growth or decline.
The series is stationary in mean.

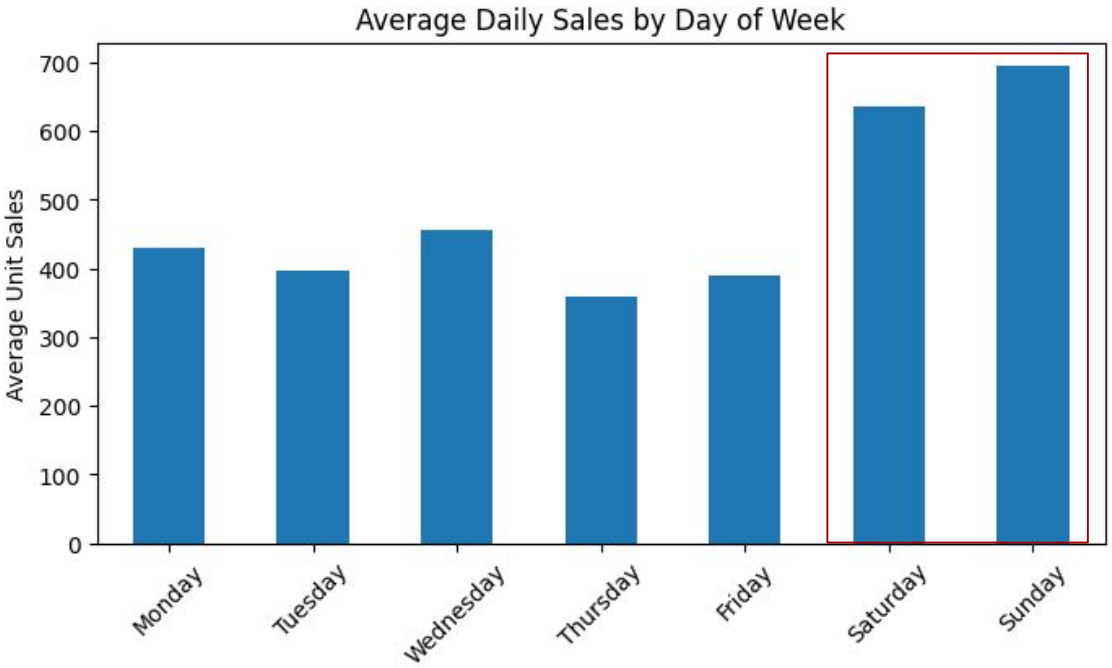
A dominant, consistent weekly pattern. The most predictable signal in the data.

Occasional spikes that no model can fully capture. A realistic ceiling on accuracy.

Weekends Drive Sales

Weekly pattern is consistent and dominant.

This weekly rhythm enables better staffing and inventory planning for high-demand days.



Feature Engineering Strategy

Feature Group	What it captures
Lag features (1, 2, 3, 7, 14 days)	Short and medium-term demand memory
Rolling statistics (7, 14, 28 days)	Local trend and volatility
Cyclical encoding (sin/cos)	Weekly rhythm without artificial discontinuity
Holiday indicators	National, Regional, Local × weekday/weekend

16 features. All derived from data structure, no arbitrary choices.

A Systematic Journey Through Model Families

All models evaluated on the same train/test split. No evaluation bias.

Evaluation metrics: MAE · RMSE · sMAPE. Primary metric: **MAE**.

📊 Statistical Models

- Naïve
- ARIMA
- SARIMA - SARIMAX
- Holt-Winters
- Prophet

🌳 Machine Learning

- Linear Regression
- Random Forest
- XGBoost

🧠 Deep Learning

- LSTM

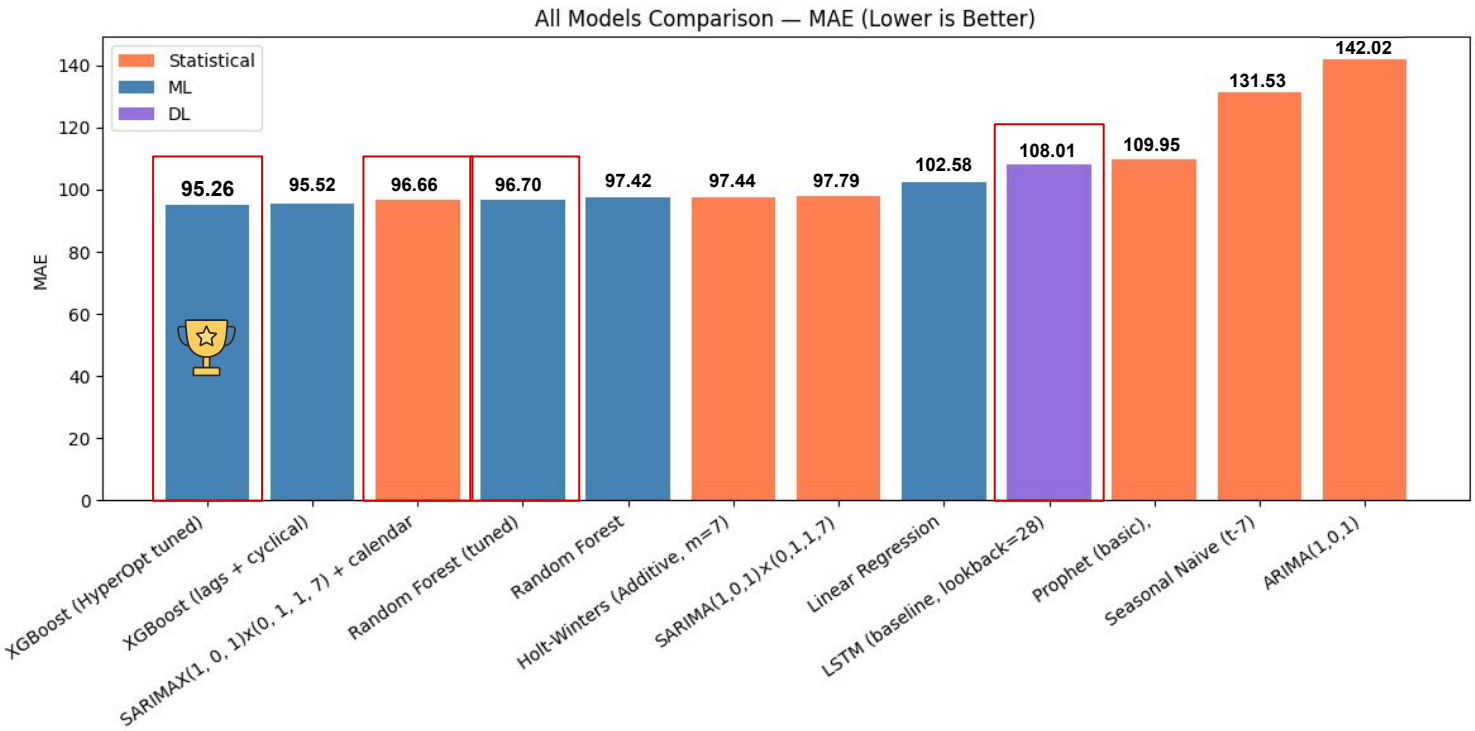
Every Decision Tracked

mlflow

			Metrics			Tags
Run Name	Duration	Models	MAE	RMSE	sMAPE	selected_candidate
XGBoost_FE_HyperOpt	1.9s	XGBoost	95.26	140.39	19.95	yes
XGBoost_FE_baseline	2.5s	XGBoost	95.52	140.94	20.04	-
SARIMAX_101_0117_calendar	2.2s	SARIMAX	96.66	143.01	20.38	-
RandomForest_tuned	1.8s	RandomForest	96.7	141.77	20.15	-
HoltWinters_additive	1.7s	Holt-Winters	97.44	143.86	20.57	-
SARIMA_101_0117	1.8s	SARIMA	97.8	145.23	20.6	-

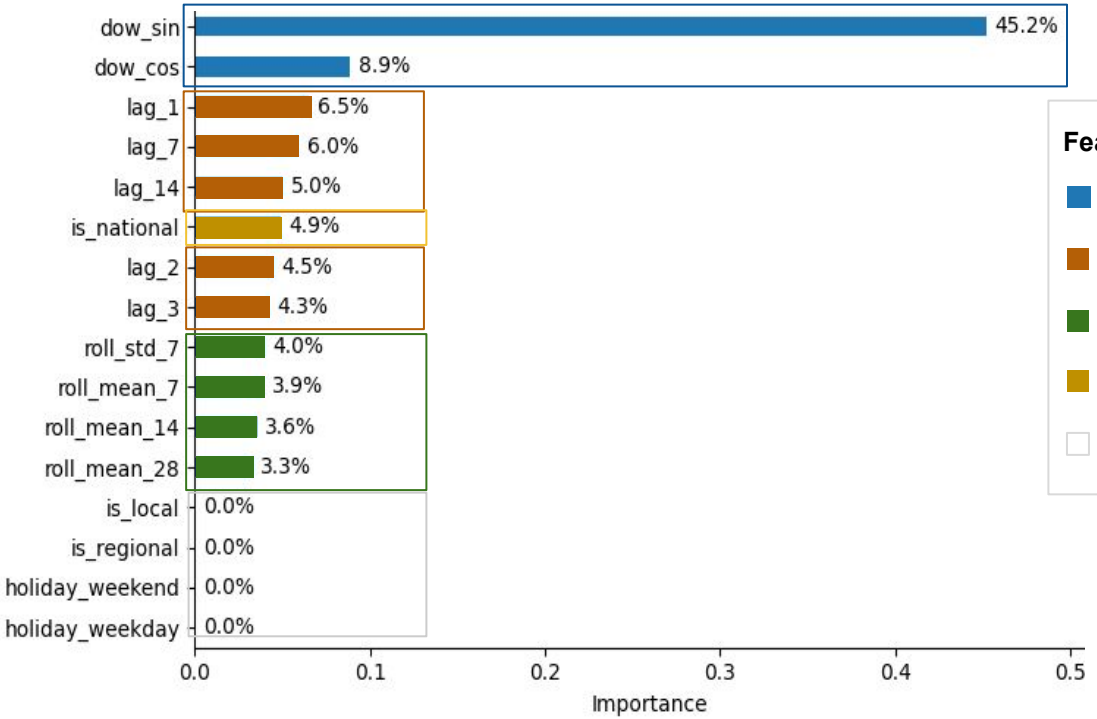
XGBoost Wins - But the Margin Tells a Story

Best model: XGBoost (HyperOpt tuned), MAE 95.26 units/day



What Actually Drives the Forecast

Champion model: XGBoost (HyperOpt tuned)

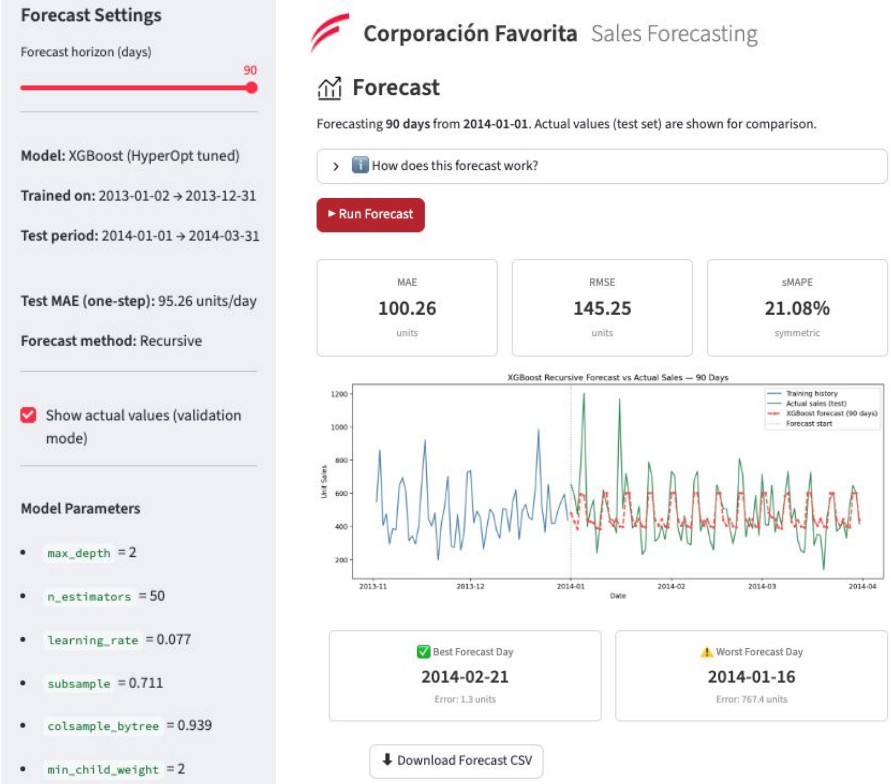


Feature Group

- **Weekly rhythm** dow_sin, dow_cos (~54%)
- **Demand memory** lag_1, lag_2, lag_3, lag_7, lag_14
- **Local volatility** roll_mean, roll_std (~15%)
- **Calendar effect** is_national (~5%)
- **No signal** is_local, is_regional, holiday_weekday, holiday_weekend (~0%)

From Model to Product

Example views from the interactive forecasting app.



Forecasting Reduces Uncertainty - It Does Not Eliminate It

- ✅ **What we can predict reliably**

  Weekly demand patterns.

  Business actions

  - Align staffing with predictable demand cycles
  - Adjust inventory orders ahead of peak days
  - Plan store logistics and supply chain flows

- ⚠️ **What remains uncertain**

  Atypical demand spikes. January 5th and 16th showed unexplained peaks no model captured.

- 📈 **What would improve accuracy**

  More years of data. A single year limits seasonality modeling and cold-start recovery.

Thank you